

Sriram Hariharan Neelakantan

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PROFESSIONAL EXPERIENCE

Techmentee Inc. , Sr. Data Scientist | Harrison, NJ

Jul 2025 – Current

- Cut average response time from 3 min to 45 sec by developing an LLM-based multi-agent intent framework (LangGraph + Step Functions) automating tier-1 FinTech chats like balance and KYC queries
- Expanded automation to nine workflows by building API-driven Action Agents (Lambda + Step Functions) for refund, payment retry, and card block, removing human review for low-risk cases
- Boosted resolution accuracy by 12% via a human-in-the-loop feedback loop retraining classifier agents and refining Bedrock prompt logic with dynamic evaluation metrics

L.B. Foster Salient Systems, Data Scientist II | Dublin, OH

Dec 2024 – Jun 2025

- Built a scalable data pipeline using Python & PySpark on Databricks with Azure Blob Storage to process 3M+ high-speed sensor data, enabling automated wheel filtering, anomaly detection, poly curve analysis, and weight calculation
- Developed a wheel health metric using double differential filtering and polynomial curve fitting with K-means to filter out 35% of jittery wheels and detect 18% more true electronic anomalies, boosting predictive accuracy
- Partnered with railway clients, operations teams, and engineering leads to gather requirements and deliver custom Power BI dashboards integrated with SQL queries from Databricks for maintenance KPIs
- Automated ADC-based sensor data validation across 16 circuits per rail, reducing misclassification by 40% and strengthening anomaly isolation and wheel health diagnostics
- Performed advanced EDA in Python and PySpark on Databricks to identify trends and anomalies, enabling proactive wheel replacement and optimized maintenance schedules

Norfolk Southern Corp, Machine Learning Intern | Atlanta, GA

Sep 2022 – Dec 2022

- Leveraged KNN, PCA, and K-means to cluster and enhance spatial analysis of 90k+ rail-road intersections, advancing infrastructure planning efforts
- Streamlined EDA using Pandas, NumPy, and Matplotlib to surface 50+ anomalies across 30k+ records, boosting clustering accuracy by 40%
- Implemented Faster R-CNN using PyTorch to detect defects in railway components, classifying 1,000+ defects from 7k undercarriage images
- Improved defect classification accuracy by 12% through advanced image processing techniques including edge detection and segmentation

Accenture, Data Analyst II | India

Mar 2020 – Jul 2021

- Engineered controlled A/B experiments and uplift analyses that identified effective campaign levers, delivering a 5.5% increase in conversions
- Deployed gradient boosting and logistic regression to segment high-risk customers, reducing early churn by 15% through targeted retention programs
- Instituted a KPI framework by converting lifecycle goals into measurable hypotheses in partnership with business teams, reducing reporting delays by 20%
- Orchestrated automated ingestion and transformation of 250k+ customer records weekly using SQL and Python, achieving 98% data quality

- Launched Tableau dashboards consolidating retention and engagement KPIs, shortening decision cycles by 30%
- Designed fraud detection models using gradient boosting and anomaly detection, identifying 22% more suspicious claim patterns and mitigating significant financial exposure

Accenture, Data Analyst I | India

Jan 2018 – Mar 2020

- Evaluated pricing strategies and outreach campaigns through causal analysis and experimentation, boosting acquisition by 8% and retention by 10%
- Designed ARIMA-based forecasting tools that detected lifecycle metric shifts, improving forecast accuracy by 12%
- Developed self-service Tableau and Power BI dashboards that standardized engagement KPIs, reducing manual reporting by 25%
- Delivered predictive insights to senior stakeholders, influencing pricing adjustments and campaign redesigns that generated a 7% ROI lift
- Reduced operational costs by 20% by identifying inefficiencies in freight distributions using logistic regression and time-series forecasting with Amazon Redshift

ACADEMIC PROJECTS

Strategic Analysis of Hospitality Data | EDA, Classification, Time Series Analysis

- Applied SVM, Logistic Regression, and XGBoost on Expedia hotel data with ARIMA-based trend analysis to model ADR and cancellation behavior

Healthcare Bot | LLM, CUDA

- Built an LLM-powered healthcare assistant using FLAN-T5, GPT-2, LSTM, and BERT, integrating user feedback to improve diagnostic accuracy and response quality

SKILLS

Programming Languages & Frameworks: Python (NumPy, Pandas, Scikit-Learn), R, PySpark, Java (REST APIs), Git, Bitbucket, JIRA, Linux / UNIX

Databases & Cloud Technologies: SQL, PostgreSQL, MySQL, MongoDB, Apache Spark, AWS EC2, AWS S3, Hadoop, Redshift

Data Analytics & Reporting: Predictive analytics, ETL, feature engineering, supervised and unsupervised ML, NLP, probability & statistics, hypothesis testing, time-series forecasting (ARIMA), Alteryx, Databricks

Data Visualization & Tools: Tableau, Excel, Matplotlib, Seaborn, Power BI, Databricks

EDUCATION

Master of Science, MS in Data Science

Sep 2021 – Dec 2023

Northeastern University, Boston, MA